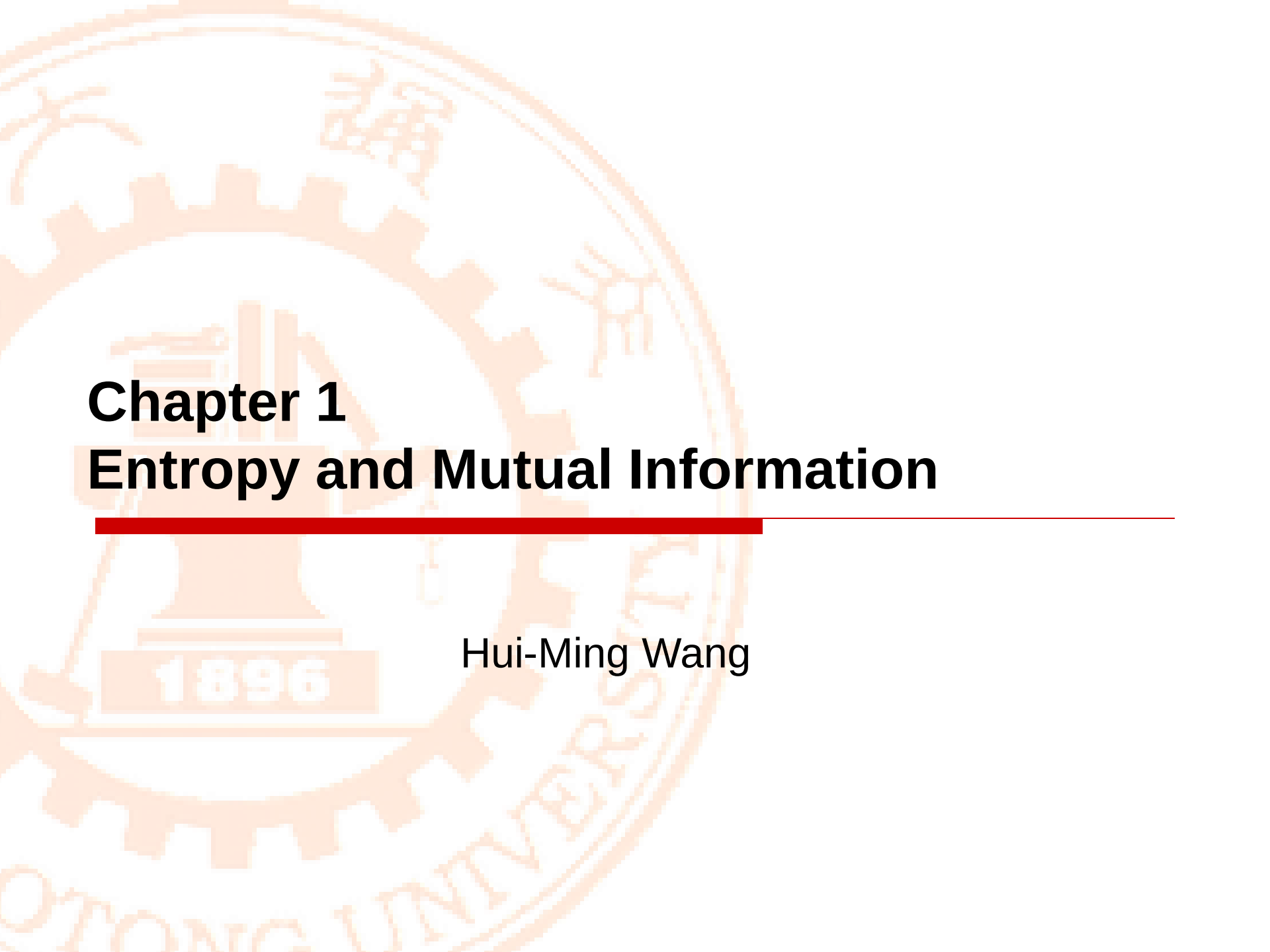


The background of the slide features a large, faint, orange-colored watermark of the Tsinghua University seal. The seal is circular and contains the university's name in both Chinese and English, along with the founding year 1896.

Chapter 1

Entropy and Mutual Information

Hui-Ming Wang



Outline

- Entropy (熵)
- Joint Entropy and Conditional Entropy (联合熵和条件熵)
- Mutual Information (互信息)
- Chain Rules (链式法则)
- Some Inequalities
 - Jensen's Inequality (杰森不等式)
 - Data-Processing Inequality (数据处理不等式)
- Summary

Entropy and Mutual Information

As we have pointed out, in Information theory we focus on the ***quantification of information***, so the first question is how to **define the amount of information**.

Measure the amount of information requires defining the information first.



Entropy

- What is information?
 - Information is the *uncertainty*. (不确定性)
 - 太阳会从东方升起；明天早上会下雨
 - 概率事件
- How to describe the uncertainty, mathematically?
 - Random variable. (随机变量)
- How to define the amount of information?
 - Entropy(熵)



Entropy

□ Entropy (Shannon Entropy)

- A measure of the **uncertainty** of a **random variable**. **All the discussions are based on the probability theory.**
- We first only consider **discrete** random variable (finite possibilities), continuous case will be considered later.
- For a discrete random variable, X , it has alphabet $\mathcal{X}$ and *probability mass function* $p(x) = \Pr\{X = x\}, x \in \mathcal{X}$.

$\log \frac{1}{p(x_i)}$ is the “information” when x_i happens

If $p(x_i)$ close to 1 $\rightarrow$ almost no information

If $p(x_i)$ close to 0 $\rightarrow$ a lot of information



Entropy

- An example
 - 32 teams are in the final of FIFA World Cup 20XX
 - Brazil, England, France, Germany, Argentina,, *China* !
 - Which will be the champion ?
 - Brazil is the champion – not so surprising – less information !
 - China is the champion !!!! – great surprise – more information !
- **Entropy** is the amount of information produced in average by a random variable.

Definition The *entropy* $H(X)$ of a discrete random variable X is defined by

$$H(X) = - \sum_{x \in \mathcal{X}} p(x) \log p(x). \quad (2.1)$$

X be a discrete random variable with alphabet $\mathcal{X}$ and *probability mass function* $p(x) = \Pr\{X = x\}$, $x \in \mathcal{X}$.



Entropy

- Expectation interpretation (期望角度的解释)

$$X \sim p(x) \quad E_p[g(X)] = \sum p(x)g(x)$$

$$H(X) = E_p\left[\log \frac{1}{p(X)}\right] = -\sum p(x)\log p(x)$$

随机变量的
函数的
期望

The expected value (**average**) of the random variable $\log \frac{1}{p(X)}$

If $\log_2 \rightarrow$ bits, if $\ln \rightarrow$ nats, $\log_2 2 = 1$ bit $\log_e 2 = \ln 2 = 0.693$ nats

1 bit = 0.693 nats; 1 nat = 1.443 bit

- ***Event that happens with probability very small contains a lot of information but do not change the entropy.***

小概率事件包含更多的信息，但是并不会改变熵的大小



Entropy

- Actually, the notion of entropy comes from **Thermo-dynamics**.

热力学

$$S = k (\ln \Omega)$$

↖ ↗

Boltzmann's constant The Number of random microstates

$$= -k (\ln 1/\Omega)$$

↖

The probability of each random microstates

$$= -k \left(\sum 1/\Omega \ln 1/\Omega \right)$$



Entropy

- **Entropy is only related to the probability mass function, but not to what the events are.**

$$\begin{bmatrix} x \\ p(x) \end{bmatrix} = \begin{bmatrix} a_1 & a_2 & a_3 \\ 1/2 & 1/3 & 1/6 \end{bmatrix}, \quad \begin{bmatrix} y \\ p(y) \end{bmatrix} = \begin{bmatrix} b_1 & b_2 & b_3 \\ 1/6 & 1/2 & 1/3 \end{bmatrix}$$

- The entropy of a fair coin toss is 1 bit (抛硬币)

- Let

$$X = \begin{cases} a & \text{with probability } \frac{1}{2}, \\ b & \text{with probability } \frac{1}{4}, \\ c & \text{with probability } \frac{1}{8}, \\ d & \text{with probability } \frac{1}{8}. \end{cases}$$

$$H(X) = -\frac{1}{2} \log \frac{1}{2} - \frac{1}{4} \log \frac{1}{4} - \frac{1}{8} \log \frac{1}{8} - \frac{1}{8} \log \frac{1}{8} = \frac{7}{4} \text{ bits.}$$



Entropy

□ Properties of entropy

- $H(X) \geq 0$ since $0 \leq p(x_i) \leq 1$

$$H(X) = - \sum_{x \in \mathcal{X}} p(x) \log p(x).$$

- $H(X) = 0$ iff $\exists x_i, p(x_i) = 1$

→ Determinant event gives no uncertainty (确定性事件没有任何不确定性)

- $H_b(X) = (\log_b a) H_a(X)$

- For a given n , the maximum $H(X)$ is equal to $\log n$, when

$$p(x_i) = \frac{1}{n}, \quad \forall i$$

we will prove this later.

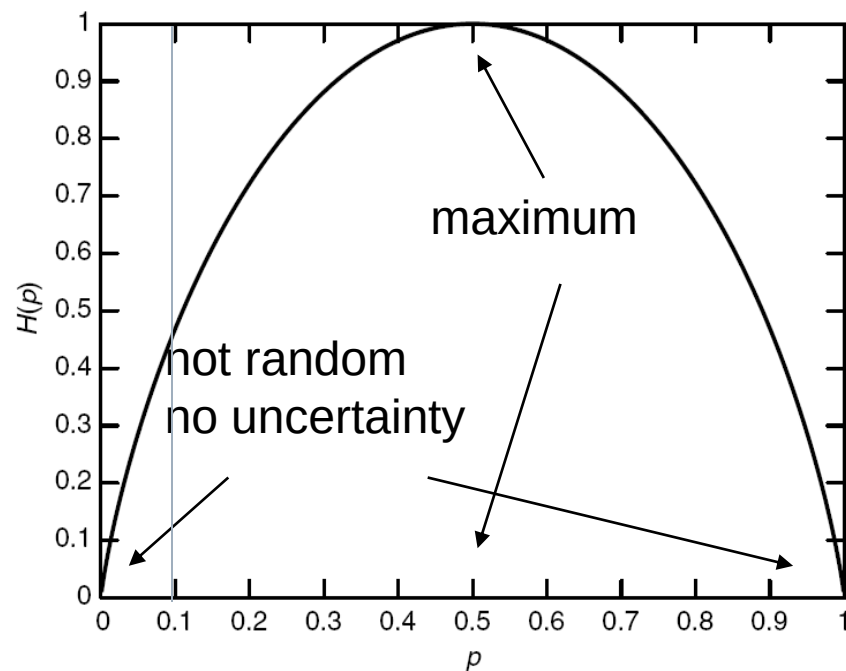
Entropy

□ Example

$$X = \begin{cases} 1 & \text{with probability } p, \\ 0 & \text{with probability } 1 - p. \end{cases}$$

$$H(p) \stackrel{\text{def}}{=} -p \log p - (1 - p) \log(1 - p)$$

$0 \log 0 = 0$
since $x \log x \rightarrow 0$ as $x \rightarrow 0$.





Joint Entropy and Conditional Entropy

□ Joint Entropy

- Extend the definition from a single random variable to a pair of random variables (从单个随机变量扩展到多个)
- Single is simple; double is trouble; triple is terrible. 😊

Definition The *joint entropy* $H(X, Y)$ of a pair of discrete random variables (X, Y) with a joint distribution $p(x, y)$ is defined as

$$H(X, Y) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(x, y),$$

which can also be expressed as $H(X, Y) = -E \log p(X, Y)$.

In this definition (X, Y) can be considered to be a single vector-valued random variable.



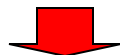
Joint Entropy and Conditional Entropy

□ Joint Entropy

$$H(X, Y) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(x, y),$$

Let (X, Y) have the following joint distribution:

$Y \backslash X$	1	2	3	4
1	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$
2	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{32}$	$\frac{1}{32}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
4	$\frac{1}{4}$	0	0	0



$\frac{1}{8}$ $\frac{1}{16}$ $\frac{1}{32}$ $\frac{1}{32}$ $\frac{1}{16}$ $\frac{1}{8}$ $\frac{1}{32}$ $\frac{1}{32}$ $\frac{1}{16}$ $\frac{1}{16}$ $\frac{1}{16}$ $\frac{1}{4}$ 0 0



Joint Entropy and Conditional Entropy

□ Conditional entropy

- Define the conditional entropy of a random variable given another as ----- the expected value of the entropies of the conditional distributions, averaged over the conditioning random variable.

Definition If $(X, Y) \sim p(x, y)$, the *conditional entropy* $H(Y|X)$ is defined as

$$\begin{aligned} H(Y|X) &= \sum_{x \in \mathcal{X}} p(x) \underline{H(Y|X = x)} \\ &= - \sum_{x \in \mathcal{X}} p(x) \sum_{y \in \mathcal{Y}} p(y|x) \log p(y|x) \\ &= - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(y|x) \\ &= -E \log p(Y|X). \end{aligned}$$

Uncertainty about Y give $X = x$

Average

Joint Entropy and Conditional Entropy

□ Conditional entropy

Let (X, Y) have the following joint distribution:

$Y \backslash X$	1	2	3	4	$p(Y)$
1	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{4}$
2	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{4}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{4}$
4	$\frac{1}{4}$	0	0	0	$\frac{1}{4}$
$p(X)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	

Marginal (边缘分布)

$$\begin{aligned}
 H(X|Y) &= \sum_{i=1}^4 p(Y=i) H(X|Y=i) \\
 &= \frac{1}{4} H\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right) + \frac{1}{4} H\left(\frac{1}{4}, \frac{1}{2}, \frac{1}{8}, \frac{1}{8}\right) \\
 &\quad + \frac{1}{4} H\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right) + \frac{1}{4} H(1, 0, 0, 0) \\
 &= \frac{1}{4} \times \frac{7}{4} + \frac{1}{4} \times \frac{7}{4} + \frac{1}{4} \times 2 + \frac{1}{4} \times 0 \\
 &= \frac{11}{8} \text{ bits.} \\
 H(Y|X) &= \frac{13}{8} \text{ bits}
 \end{aligned}$$

$\left. \begin{array}{l} \frac{11}{8} \text{ bits.} \\ \frac{13}{8} \text{ bits} \end{array} \right\} H(Y|X) \neq H(X|Y)$



Joint Entropy and Conditional Entropy

□ Chain rule

- The entropy of a pair of random variables is the entropy of one plus the conditional entropy of the other.
- **Joint entropy = entropy + conditional entropy**

Theorem 2.2.1 (*Chain rule*)

$$H(X, Y) = H(X) + H(Y|X).$$

joint probability = marginal probability X conditional probability

联合概率=边缘概率X条件概率

$$p(x, y) = p(x)p(y|x)$$

贝叶斯公式



Joint Entropy and Conditional Entropy

Proof

$$\begin{aligned} H(X, Y) &= - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(x, y) \\ &= - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(x) p(y|x) \\ &= - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(x) - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(y|x) \\ &= - \sum_{x \in \mathcal{X}} p(x) \log p(x) - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(y|x) \\ &= H(X) + H(Y|X). \end{aligned}$$

■ Corollary
Conditional entropy chain rule

$$H(X, Y|Z) = H(X|Z) + H(Y|X, Z).$$



Mutual Information

□ Mutual information

- A measure of the amount of information that one random variable contains about another random variable.

$$I(X; Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} = E_{p(x, y)} \log \frac{p(X, Y)}{p(X)p(Y)}$$

$Y \backslash X$	1	2	3	4	$p(Y)$
1	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{4}$
2	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{4}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{4}$
4	$\frac{1}{4}$	0	0	0	$\frac{1}{4}$
$p(X)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	

$$I(X; Y) = 0.375 \text{ bit.}$$

Marginal (边缘分布)

Mutual Information

□ Mutual information

$$\begin{aligned}
 I(X; Y) &= \sum_{x,y} p(x, y) \log \frac{p(x|y)}{p(x)} = - \sum_{x,y} p(x, y) \log p(x) + \sum_{x,y} p(x, y) \log p(x|y) \\
 &= \underbrace{- \sum_x p(x) \log p(x)}_{H(X)} - \underbrace{\left(- \sum_{x,y} p(x, y) \log p(x|y) \right)}_{H(X|Y)}
 \end{aligned}$$

- It is the reduction in the uncertainty of one random variable due to the knowledge of the other.

- If we just look at X , the uncertainty is $H(X)$
- Given Y , uncertainty about X , $H(X|Y)$

Reduction of
uncertainty
 $H(X) - H(X|Y)$



Mutual Information

- By symmetry, it also follows that

$$I(X; Y) = H(Y) - H(Y|X).$$

- The ***mutual information*** of a random variable with itself is the ***entropy*** of the random variable.

随机变量和自己的互信息就是这个随机变量的熵

$$I(X; X) = H(X) - H(X|X) = H(X).$$

- Entropy is sometimes referred to as ***self-information*** (自信息).

Mutual Information

□ Relationship (Venn diagram)

Theorem 2.4.1 (*Mutual information and entropy*)

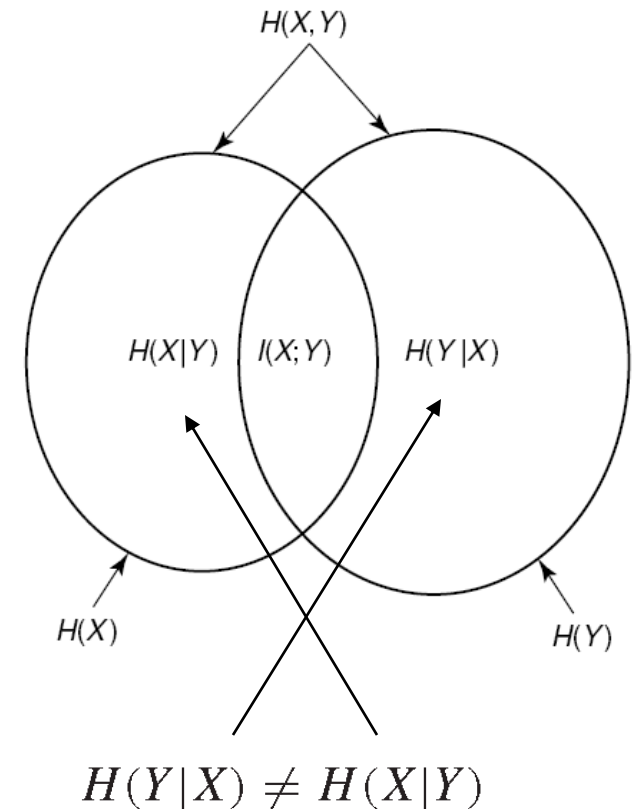
$$I(X; Y) = H(X) - H(X|Y)$$

$$I(X; Y) = H(Y) - H(Y|X)$$

$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

$$I(X; Y) = I(Y; X)$$

$$I(X; X) = H(X).$$





Mutual Information

□ Examples

$Y \backslash X$	1	2	3	4	$p(Y)$
1	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{4}$
2	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{4}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{4}$
4	$\frac{1}{4}$	0	0	0	$\frac{1}{4}$
$p(X)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	

$$H(X) = \frac{7}{4} \text{ bits}$$

$$H(Y) = 2 \text{ bits}$$

$$H(Y|X) = \frac{13}{8} \text{ bits}$$

$$\begin{aligned} I(X; Y) &= H(X) - H(X|Y) \\ &= H(Y) - H(Y|X) \\ &= 0.375 \text{ bit.} \end{aligned}$$

Mutual Information

□ Conditional mutual information

Definition The *conditional mutual information* of random variables X and Y given Z is defined by

$$I(X; Y|Z) = H(X|Z) - H(X|Y, Z)$$
$$= E_{p(x,y,z)} \log \frac{p(X, Y|Z)}{p(X|Z)p(Y|Z)}.$$

$I(X; Y) = H(X) - H(X|Y)$

 $= E_{p(x,y)} \log \frac{p(X, Y)}{p(X)p(Y)}$

How to prove it mathematically?



Chain Rules

□ Chain rule for entropy (熵的链式法则)

Theorem 2.5.1 (Chain rule for entropy) Let $X_1, X_2, \dots, X_n$ be drawn according to $p(x_1, x_2, \dots, x_n)$. Then

$$H(X_1, X_2, \dots, X_n) = \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1).$$

$$H(X_1, X_2, \dots, X_n) = H(X_1) + H(X_2 | X_1) + \dots + H(X_n | X_{n-1}, \dots, X_1)$$

Proof:

$$H(X_1, X_2) = H(X_1) + H(X_2 | X_1),$$

$$H(X_1, X_2, X_3) = H(X_1) + H(X_2, X_3 | X_1)$$

$$= H(X_1) + H(X_2 | X_1) + H(X_3 | X_2, X_1),$$

$\vdots$

$$H(X_1, X_2, \dots, X_n) = H(X_1) + H(X_2 | X_1) + \dots + H(X_n | X_{n-1}, \dots, X_1)$$

Chain rule

Conditional
chain rule



Chain Rules

□ Chain rule for information (互信息的链式法则)

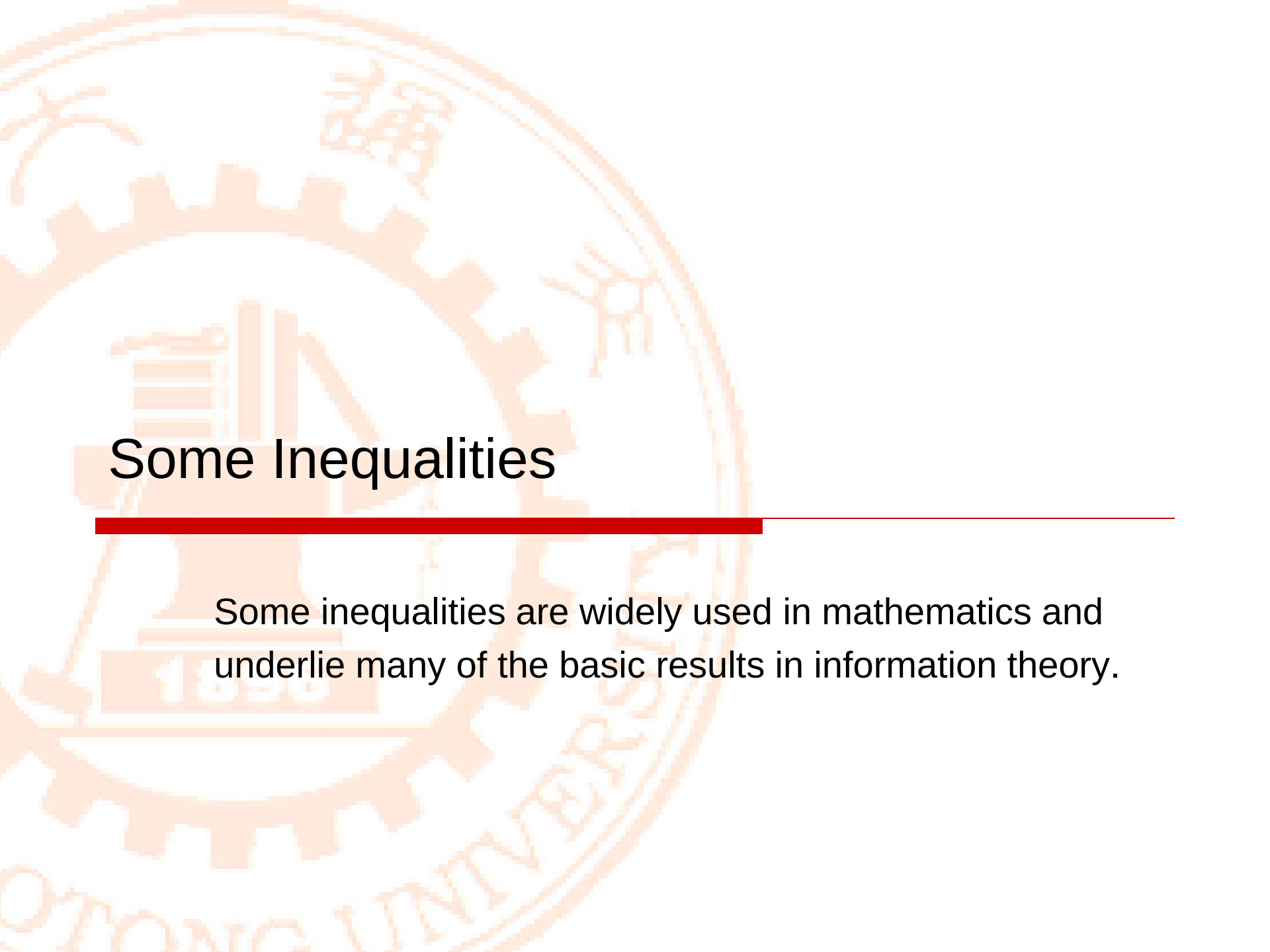
Theorem 2.5.2 (*Chain rule for information*)

$$I(X_1, X_2, \dots, X_n; Y) = \sum_{i=1}^n I(X_i; Y | X_{i-1}, X_{i-2}, \dots, X_1).$$

Proof

$$\begin{aligned} I(X_1, X_2, \dots, X_n; Y) &= \underbrace{H(X_1, X_2, \dots, X_n)}_{\text{C.R. for entropy}} - \underbrace{H(X_1, X_2, \dots, X_n | Y)}_{\text{Conditional chain rule}} \\ &= \sum_{i=1}^n \underbrace{H(X_i | X_{i-1}, \dots, X_1)}_{\text{C.R. for entropy}} - \sum_{i=1}^n \underbrace{H(X_i | X_{i-1}, \dots, X_1, Y)}_{\text{Conditional chain rule}} \\ &= \sum_{i=1}^n I(X_i; Y | X_1, X_2, \dots, X_{i-1}). \quad \square \end{aligned}$$

Conditional mutual information

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Some Inequalities

Some inequalities are widely used in mathematics and underlie many of the basic results in information theory.

Convexity

□ Convex function

Definition A function $f(x)$ is said to be *convex* over an interval (a, b) if for every $x_1, x_2 \in (a, b)$ and $0 \leq \lambda \leq 1$,

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2).$$

A function f is said to be *strictly convex* if equality holds only if $\lambda = 0$ or $\lambda = 1$.

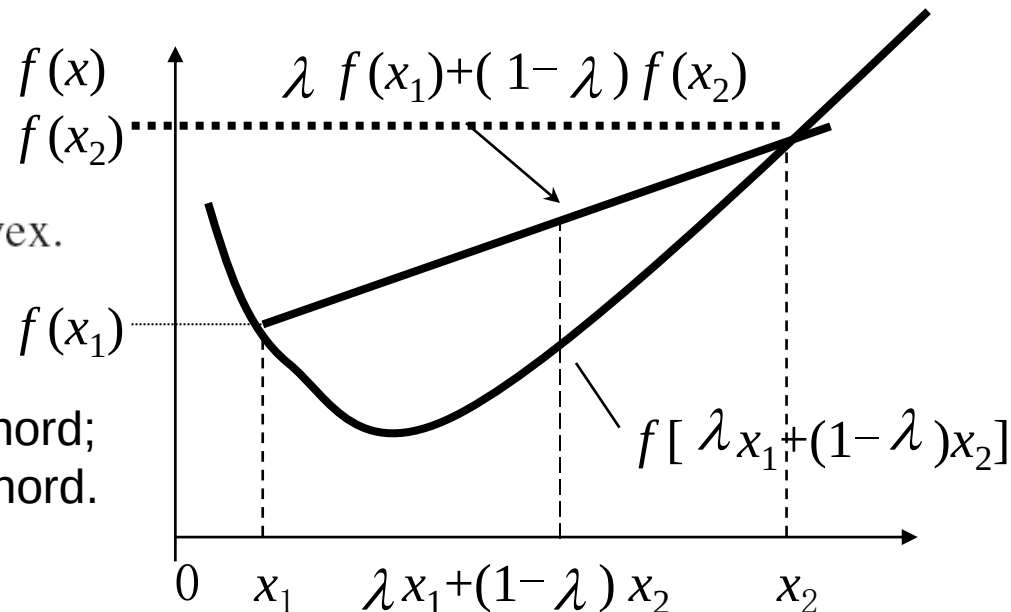
Concave function

A function f is *concave* if $-f$ is convex.

A function is

Convex: if it always lies **below** any chord;

Concave: if it always lies **above** any chord.



Convexity

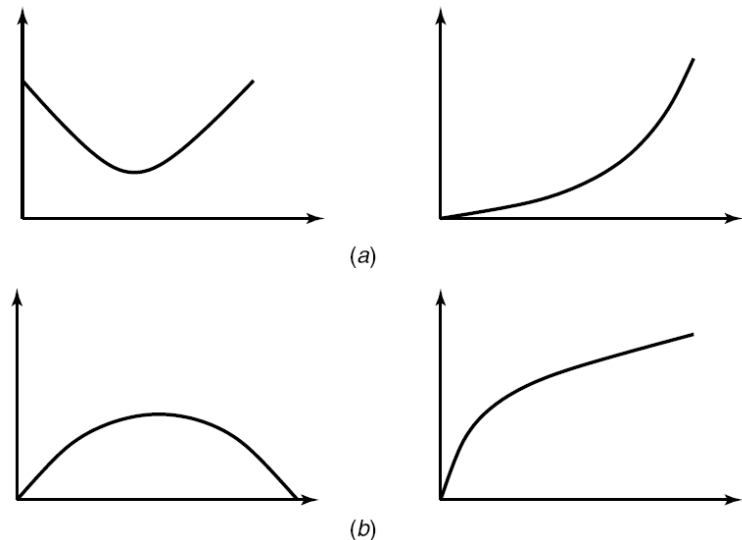
□ Examples

convex functions (凸函数)

x^2 , $|x|$, e^x , $x \log x$ (for $x \geq 0$),

concave functions (凹函数)

$\log x$ $\sqrt{x}$ $x \geq 0$.



Convexity underlies many of the basic properties of information-theoretic quantities such as entropy and mutual information.

Theorem 2.6.1 *If the function f has a second derivative that is non-negative (positive) over an interval, the function is convex (strictly convex) over that interval.*



Jensen's inequality

□ Jensen's inequality

Theorem 2.6.2 (*Jensen's inequality*) If f is a convex function and X is a random variable,

$$Ef(X) \geq f(EX).$$

Moreover, if f is strictly convex, the equality implies that $X = EX$ with probability 1 (i.e., X is a constant).

where $EX = \sum_{x \in \mathcal{X}} p(x)x$ denotes the expectation.

The expectation of a convex function of a random variable is **larger** or equal to the convex function of the expectation of the random variable.

“凸函数的期望大于等于期望的凸函数”

If the function is a **concave** function, then we have:

The expectation of a concave function of a random variable is **less** or equal to the concave function of the expectation of the random variable.

“凹函数的期望小于等于期望的凹函数”



Jensen's inequality

□ Proof: **Mathematical induction** (数学归纳法)

For a two-mass-point distribution

$$p_1 f(x_1) + p_2 f(x_2) \geq f(p_1 x_1 + p_2 x_2), \quad \text{convex}$$

Suppose that the theorem is true for distributions with $k - 1$ mass points.

$$\begin{aligned} \sum_{i=1}^k p_i f(x_i) &= p_k f(x_k) + (1 - p_k) \sum_{i=1}^{k-1} p'_i f(x_i) \quad \leftarrow p'_i = p_i / (1 - p_k) \\ &\geq p_k f(x_k) + (1 - p_k) f\left(\sum_{i=1}^{k-1} p'_i x_i\right) \quad \leftarrow \sum_{i=1}^{k-1} p'_i = 1 \\ &\geq f\left(p_k x_k + (1 - p_k) \sum_{i=1}^{k-1} p'_i x_i\right) \\ &= f\left(\sum_{i=1}^k p_i x_i\right), \end{aligned}$$

Jensen's inequality

□ Application of Jensen's inequality

■ Non-negativity of mutual information (互信息的非负性)

(Nonnegativity of mutual information) For any two random variables, X, Y ,

$$I(X; Y) \geq 0,$$

with equality if and only if X and Y are independent.

Proof: $-I(X; Y) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$

$$= -E_{p(x, y)} \log \frac{p(X, Y)}{p(X)p(Y)}$$

$$= E_{p(x, y)} \log \frac{p(X)p(Y)}{p(X, Y)}$$

Jensen's
inequality
concave function



$$\leq \log \sum_x \sum_y p(X, Y) \frac{p(X)p(Y)}{p(X, Y)} = \log \sum_x \sum_y p(X)p(Y) = 0$$

Equation holds when

$$\frac{p(x)p(y)}{p(x, y)} = c, \quad \forall x, y$$

$$\rightarrow p(x, y) = p(x)p(y)$$



Jensen's inequality

- Conditioning reduces entropy (条件降低熵)

Theorem 2.6.5 (*Conditioning reduces entropy*)

$$H(X|Y) \leq H(X)$$

with equality if and only if X and Y are independent.

Proof: $0 \leq I(X; Y) = H(X) - H(X|Y)$.

Intuitively, the theorem says that knowing another random variable Y can only reduce the uncertainty in X .

Note that this is true only on the **average**.

$$H(X) = H\left(\frac{1}{8}, \frac{7}{8}\right) = 0.544 \text{ bit},$$

$$H(X|Y = 1) = 0 \quad H(X|Y = 2) = 1$$

$$H(X|Y) = \frac{3}{4}H(X|Y = 1) + \frac{1}{4}H(X|Y = 2) = 0.25$$

Y \ X	1	2	
	0	$\frac{3}{4}$	$\frac{3}{4}$
2	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$
	$\frac{1}{8}$	$\frac{7}{8}$	



Jensen's inequality

■ Bound on joint entropy (联合熵的上界)

Theorem 2.6.6 (*Independence bound on entropy*)
 $X_1, X_2, \dots, X_n$ be drawn according to $p(x_1, x_2, \dots, x_n)$. Then

Let

$$H(X_1, X_2, \dots, X_n) \leq \sum_{i=1}^n H(X_i)$$

with equality if and only if the X_i are independent.

Proof: By the chain rule for entropies,

$$H(X_1, X_2, \dots, X_n) = \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1)$$

We have equality if and only if
 X_i 's are independent.

$$\leq \sum_{i=1}^n H(X_i),$$



Data-processing inequality

□ Markov chain (马尔可夫链)

Definition Random variables X, Y, Z are said to form a Markov chain in that order (denoted by $X \rightarrow Y \rightarrow Z$) if the conditional distribution of Z depends only on Y and is conditionally independent of X . Specifically, X, Y , and Z form a Markov chain $X \rightarrow Y \rightarrow Z$ if the joint probability mass function can be written as

$$p(x, y, z) = p(x)p(y|x)p(z|y).$$

- $X \rightarrow Y \rightarrow Z$ if and only if X and Z are conditionally independent given Y .

$$p(x, z|y) = \frac{p(x, y, z)}{p(y)} = \frac{p(x, y)p(z|y)}{p(y)} = p(x|y)p(z|y).$$

- $X \leftrightarrow Y \leftrightarrow Z$.
- If $Z = f(Y)$, then $X \rightarrow Y \rightarrow Z$.



Data-processing inequality

□ Data-processing inequality (数据处理不等式)

Theorem 2.8.1 (*Data-processing inequality*) If $X \rightarrow Y \rightarrow Z$, then $I(X; Y) \geq I(X; Z)$.

Proof: By the chain rule, we can expand mutual information in two different ways:

$$\begin{aligned} I(X; Y, Z) &= I(X; Z) + I(X; Y|Z) \\ &= I(X; Y) + I(X; Z|Y). \end{aligned}$$

Since X and Z are conditionally independent given Y , we have $I(X; Z|Y) = 0$. Since $I(X; Y|Z) \geq 0$, we have

$$I(X; Y) \geq I(X; Z).$$

We have equality if and only if $I(X; Y|Z) = 0$ (i.e., $X \rightarrow Z \rightarrow Y$ forms a Markov chain).

The data-processing inequality shows that no clever manipulation of the data can improve the inferences that can be made from the original data.



Data-processing inequality

- Conditioning does not always reduce mutual information

Corollary If $X \rightarrow Y \rightarrow Z$, then $I(X; Y|Z) \leq I(X; Y)$.

Proof:
$$I(X; Y, Z) = I(X; Z) + I(X; Y|Z)$$
$$= I(X; Y) + I(X; Z|Y).$$

Conditioning reduces entropy.

Conditioning does not always reduce mutual information.

满足马尔科夫链时，才降低

It's possible that $I(X; Y|Z) > I(X; Y)$ when X , Y , and Z do not form a Markov chain. For example, let X and Y be independent fair binary random variables, and let $Z = X + Y$. Then $I(X; Y) = 0$, but $I(X; Y|Z) = H(X|Z) - H(X|Y, Z) = H(X|Z) = P(Z = 1)H(X|Z = 1) = \frac{1}{2}$ bit.

Conditional mutual information	$\frac{1}{2}$	1
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□ The entropy of a random variable is a concave function.

Proof: X $P = \{p_1, p_2, \dots, p_n\}$

$$H(X) = -\sum_{i=1}^n p_i \log p_i$$

$$f(p) = -p \log p \quad 0 < p \leq 1 \quad \longrightarrow \quad f(p) = -p \cdot \log(e) \cdot \ln(p)$$

$$f'(p) = (-\log e) \left(\ln p + p \cdot \frac{1}{p} \right) = (-\log e)(\ln p + 1);$$

$$f''(p) = (-\log e) \cdot \frac{1}{p} < 0 \quad f(p) \text{ is a concave function}$$

$$H(X) = -\sum_{i=1}^n p_i \log p_i = \sum_{i=1}^n f(p_i)$$



Theorem 2.7.4 Let $(X, Y) \sim p(x, y) = p(x)p(y|x)$. The mutual information $I(X; Y)$ is a concave function of $p(x)$ for fixed $p(y|x)$

Proof:

$$I(X; Y) = \sum_{x, y} p(x) p(y|x) \log \frac{p(y|x)}{\sum_x p(x) p(y|x)}$$
$$I(X; Y) = f(p(x))$$

Two distributions $p_1(x)$ $p_2(x)$

$$p_1(x, y) = p_1(x) p(y|x) \quad p_1(y) = \sum_x p_1(x, y)$$

$$p_2(x, y) = p_2(x) p(y|x) \quad p_2(y) = \sum_x p_2(x, y)$$

Define a new distribution $p_3(x) = \theta p_1(x) + (1 - \theta) p_2(x)$, $0 < \theta < 1$

$$p_3(x, y) = p_3(x) p(y|x) = \theta p_1(x, y) + (1 - \theta) p_2(x, y)$$



According to definition of concave function, we have to prove

$$\theta f(p_1(x)) + (1-\theta)f(p_2(x)) \leq f[\theta p_1(x) + (1-\theta)p_2(x)]$$

$$\begin{aligned} & \theta f(p_1(x)) + (1-\theta)f(p_2(x)) - f(\theta p_1(x) + (1-\theta)p_2(x)) \\ &= \theta \sum_{x,y} p_1(x) p(y|x) \log \frac{p(y|x)}{p_1(y)} + (1-\theta) \sum_{x,y} p_2(x) p(y|x) \log \frac{p(y|x)}{p_2(y)} \\ & \quad - \sum_{x,y} (\theta p_1(x) + (1-\theta)p_2(x)) p(y|x) \log \frac{p(y|x)}{p_3(y)} \\ &= \theta \sum_{x,y} p_1(x, y) \log \frac{p_3(y)}{p_1(y)} + (1-\theta) \sum_{x,y} p_2(x, y) \log \frac{p_3(y)}{p_2(y)} \end{aligned}$$



$$\theta \sum_{x,y} p_1(x,y) \log \frac{p_3(y)}{p_1(y)} \leq \log \left[\sum_{x,y} p_1(x,y) \frac{p_3(y)}{p_1(y)} \right]$$

Jensen's
inequality
concave function

$$= \log \left[\sum_y \left(\frac{p_3(y)}{p_1(y)} \sum_x p_1(x,y) \right) \right] = \log \sum_y p_3(y) = 0$$

$$\sum_{x,y} p_2(x,y) \log \frac{p_3(y)}{p_2(y)} \leq 0$$

$$\theta f(p_1(x)) + (1-\theta) f(p_2(x)) - f(\theta p_1(x) + (1-\theta) p_2(x)) \leq 0$$

$$\rightarrow \theta f(p_1(x)) + (1-\theta) f(p_2(x)) \leq f[\theta p_1(x) + (1-\theta) p_2(x)]$$



□ Another proof

$$I(X; Y) = H(Y) - H(Y|X) = H(Y) - \sum_x p(x) H(Y|X = x).$$

If $p(y|x)$ is fixed, then $p(y)$ is a linear function of $p(x)$

$$p(y) = \sum_x p(x, y) = \sum_x p(x) p(y|x)$$

$H(Y) = -\sum_{i=1}^n p(y_i) \log p(y_i)$ is a concave function of $p(y)$,

is also a concave function of $p(x)$

$f(g(x))$ composite function

$-\sum_x p(x) H(Y|X = x)$ is a linear function of $p(x)$

Hence, the difference is a concave function of $p(x)$

复合函数
线性函数的凹函
数还是凹函数

—凹函数和线性函数的差还是凹函数

Summary

- **Entropy** is a measure of the uncertainty of a *random variable*.

$$H(X) = -\sum_{x \in \mathcal{X}} p(x) \log p(x).$$

- **Properties**

- Non-negativity $H(X) \geq 0.$

- Unit Transform $H_b(X) = (\log_b a) H_a(X).$

- Conditioning reduces entropy $H(X|Y) \leq H(X)$

- Joint reduces entropy $H(X_1, X_2, \dots, X_n) \leq \sum_{i=1}^n H(X_i),$

- Uniform distribution maximize the entropy $H(X) \leq \log |\mathcal{X}|$

- Joint Entropy $H(X, Y) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(x, y).$

- Conditional entropy $H(Y|X) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(y|x)$

- Chain rule $H(X, Y) = H(X) + H(Y|X).$

Summary

- **Mutual information** is a measure of the amount of information that one random variable contains about another random variable.

$$I(X; Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}.$$

- **Conditional Mutual information** $I(X; Y|Z) = H(X|Z) - H(X|Y, Z)$

- **Relationship** $= E_{p(x,y,z)} \log \frac{p(X, Y|Z)}{p(X|Z)p(Y|Z)}.$

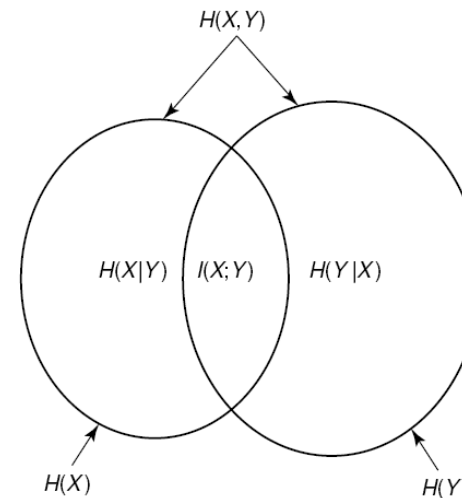
$$I(X; Y) = H(X) - H(X|Y)$$

$$I(X; Y) = H(Y) - H(Y|X)$$

$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

$$I(X; Y) = I(Y; X)$$

$$I(X; X) = H(X).$$





Summary

□ Chain rules

■ Entropy $H(X_1, X_2, \dots, X_n) = \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1).$

■ Mutual information

$$I(X_1, X_2, \dots, X_n; Y) = \sum_{i=1}^n I(X_i; Y | X_1, X_2, \dots, X_{i-1}).$$

□ Convex function and Jensen's inequality

If f is a convex function, then $Ef(X) \geq f(EX).$

□ Markov chain and Data-processing inequality

If $X \rightarrow Y \rightarrow Z$, $I(X; \bar{Y}) \geq I(X; \bar{Z}).$

The entropy of a random variable is a concave function.

The mutual information $I(X; Y)$ is a concave function of $p(x)$
for fixed $p(y|x)$